

# Gene Pulser® Electroprotocols

\* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

|                     |                                                                            |                                 |                                                                                                                    |
|---------------------|----------------------------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Cell Type</b>    | Mammalian, adherent, suspension                                            | <b>Molecules Electroporated</b> | DNA: circular plasmid recombinants (pMAM (8.8 kB + insert), pATH (9.0 kB) pGem (4.0 kB) pMSG (8.8 kB) pM6 (9.7 kB) |
| <b>Species Used</b> | Human, HeLa, cervical (C41); Human epithelial cells; Monkey, Vero, kidney. |                                 |                                                                                                                    |

## Before the Pulse

|                           |                                                                              |                                |              |
|---------------------------|------------------------------------------------------------------------------|--------------------------------|--------------|
| <b>Cell Growth Medium</b> | DMEM, MEM, M199; (10% Fetal Bovine Serum + 1% antibiotic) (GIBCO/BRL, Sigma) | <b>Growth Phase at Harvest</b> | Early log    |
|                           |                                                                              | <b>Pre-pulse Incubation</b>    | Ice, 10 min. |

**Wash Solution** Phosphate Buffered Saline (cold)

## The Pulse

**Instruments Used** Gene Pulser® apparatus & Capacitance

|                                    |                                    |                       |                           |
|------------------------------------|------------------------------------|-----------------------|---------------------------|
| <b>Electroporation Temperature</b> | Chilled                            | <b>Cuvette Gap</b>    | 0.4 cm                    |
| <b>Electroporation Medium*</b>     | Same as growth media               | <b>Voltage</b>        | 0.250 kV                  |
| <b>Cell Density</b>                | 3 x 10 <sup>6</sup> (6) cells / ml | <b>Field Strength</b> | 0.625 kV/cm               |
| <b>Volume of Cells</b>             | 400 µl                             | <b>Capacitor</b>      | 500 µF                    |
| <b>DNA Concentration</b>           | 10 to 100 µg / sample              | <b>Resistor</b>       | (Pulse Controller) Ω none |
| <b>DNA Resuspension Buffer</b>     | Not given                          | <b>Time Constant</b>  | 20 to 25 msec             |
| <b>Volume of DNA</b>               | 1 µl to 50 µl                      |                       |                           |

## After the Pulse

**Outgrowth Medium** Same as growth media

## Relevant Publications and/or Comments

**Note:** exponential values designated in parentheses.

|                                       |                                                                                    |
|---------------------------------------|------------------------------------------------------------------------------------|
| <b>Outgrowth Temperature</b>          | Brief ice, then 37°C                                                               |
| <b>Length of Incubation</b>           | 16 to 48 hours prior to selection                                                  |
| <b>Selection Method or Assay Used</b> | G418                                                                               |
| <b>Electroporation Efficiency</b>     | 1 x 10 <sup>-3</sup> to 10 <sup>-9</sup> depending on cell type and plasmids used. |
| <b>Per Cent Survival</b>              | 50%                                                                                |

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158